

L I N G U A F O R G E



Master Plan · Project Origami

"Grammar that unfolds in the mind."

□LinguaForge Grammar Engine —

Master Plan v1.0

Codename: *Project ORIGAMI — "Grammar that unfolds in the mind."* **Owner:** Chief Product Officer, LinguaForge **Status:** Production blueprint — ready for sprint planning **Target Audience:** Tech professionals (developers, IT, corporate engineers) **Pedagogical Foundation:** Cognitive Load Theory · Dual-Coding Theory · Spaced Repetition · Productive Failure

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1. PHILOSOPHY

1.1 The Core Insight

Traditional grammar teaching fails because it treats language as **rules to memorize**. Real fluency comes from **patterns the brain pre-consciously recognizes**.

Our engine transforms grammar from an *abstract rule system* into a *visible, manipulable, almost physical thing* — like watching gears mesh, water flow, or origami unfold.

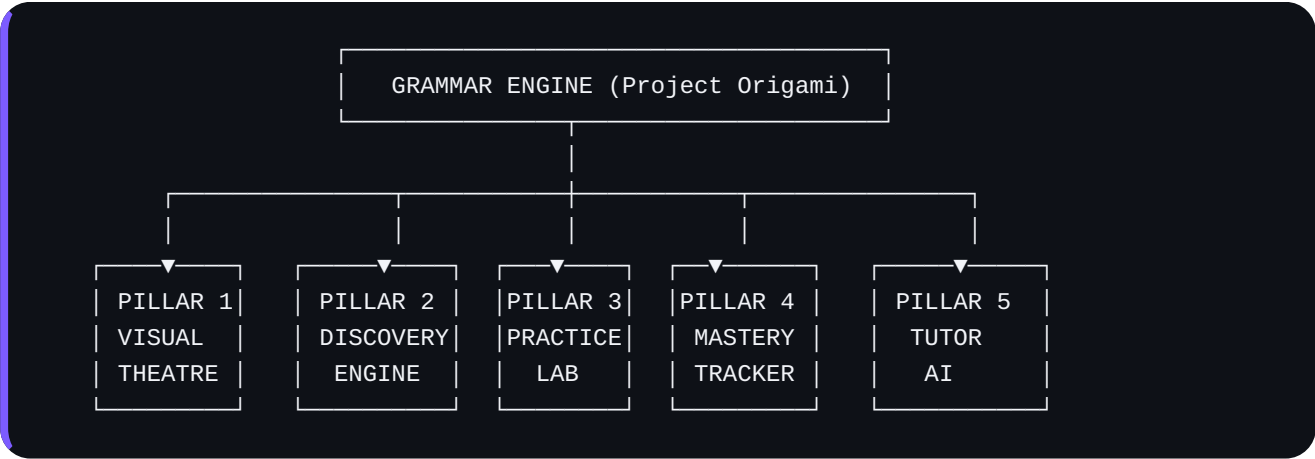
1.2 The Five Learning Laws (Non-Negotiable)

#	Law	Implementation Impact
1	Show, never tell first.	Every concept opens with a 3D animation, then explanation comes
2	One concept per breath.	Max 1 new grammar atom per screen. No cognitive overload.
3	Make the invisible visible.	Time, agency, conditionality must be <i>drawn</i> — never just described
4	Failure is the teacher.	Wrong answers trigger micro-lessons, not red Xs
5	Connect to their world.	All examples are workplace-grade: sprints, PRs, standups, interviews

1.3 Pedagogical Frameworks Applied

Framework	What It Means	How We Use It
Cognitive Load Theory (Sweller)	Working memory holds ~4 items	Each screen has ≤ 4 visual elements
Dual-Coding Theory (Paivio)	Visual + verbal = 2× retention	Every concept has animation + audio + text
Spaced Repetition (Ebbinghaus)	Memory curves predict forgetting	SuperMemo SM-2 algorithm for re-tests
Productive Failure (Kapur)	Struggle before instruction = deeper learning	Quiz BEFORE lesson on review days
Comprehensible Input + 1 (Krashen)	Slightly above current level	AI adjusts difficulty per user
Generation Effect	Creating > recognizing	70% of exercises require user to generate, not pick

2. THE 5-PILLAR ARCHITECTURE



Pillar	What It Does	Why It Matters
1. Visual Theatre	Animated 3D representations of every concept	Makes abstract visible (dual-coding)
2. Discovery Engine	Guided "aha!" moments before rules	Productive failure → deeper encoding
3. Practice Lab	14 exercise types, drag/build/speak/type	Generation effect → active recall
4. Mastery Tracker	SRS-driven review queue + skill graph	Long-term retention guaranteed
5. Tutor AI	Gemini-powered "Why am I wrong?" companion	Mistake → micro-lesson in 2 seconds

3. THE 12 GRAMMAR MODULES (Curriculum Map)

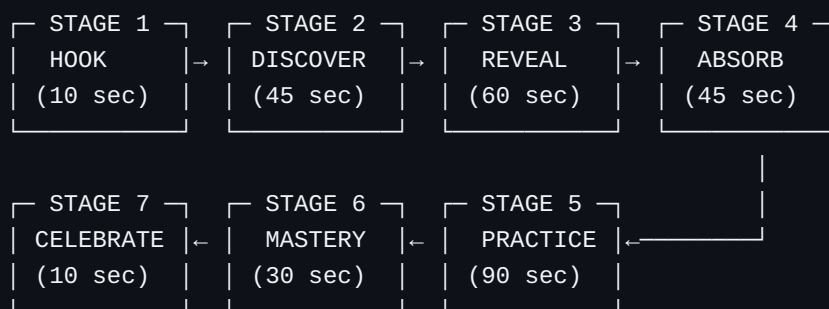
Sequenced by **frequency of use in tech workplaces**, not traditional grammar-book order.

#	Module	Why Tech Professionals Need It	Difficulty	Lessons
1	The 12 Tenses (Time Navigator)	"I deployed", "I have been deploying", "I will have deployed" — all daily	★★★	18
2	Articles (a / an / the / Ø)	#1 mistake for Indian/Asian devs — "I sent you <i>the</i> PR"	★★	6
3	Prepositions (in / on / at / by)	"Merged on Friday", "by EOD", "in production"	★★	8
4	Modals (can/could/should/must/might)	Politeness in PRs, code reviews, escalations	★★★	10
5	Active vs Passive Voice	Standups vs documentation register	★★	5
6	Conditionals (0/1/2/3/ Mixed)	Hypotheticals: "If we'd shipped earlier..."	★★★★★	8
7	Reported Speech	"He said the bug was fixed" — meetings, escalations	★★★	6
8	Subject-Verb Agreement	"The team <i>is/are</i> " — collective nouns	★★	4
9	Phrasal Verbs (tech-specific)	"Roll out", "spin up", "tear down", "ramp up"	★★★	8
10	Connectors & Cohesion	"However", "Moreover", "Consequently" — exec emails	★★	6
11	Question Formation	Interview answers, clarifications, standups	★★	5
12	Punctuation & Register	Slack vs email vs spec doc — code-switching	★★	4

Total: 88 micro-lessons · ~6 hours of pure content · Each lesson 4–6 min

4. LESSON ANATOMY — THE 7-STAGE FLOW

Every single grammar lesson follows this exact rhythm. Predictability = comfort = focus.



Stage	Duration	What Happens	UI Element	Cognitive Goal
1. HOOK	10s	A real workplace scenario appears: <i>"Your manager asks: 'Have you pushed the fix yet?' — what does she actually want to know?"</i>	Hero card with subtle 3D phone/chat mockup	Trigger curiosity
2. DISCOVER	45s	2 example sentences appear. User taps the difference. No explanation yet.	Side-by-side cards with tappable hotspots	Productive failure
3. REVEAL	60s	The 3D Tense Map / Sentence Diagram animates in, showing the underlying pattern	Signature animation (see Section 5–9)	Visual encoding
4. ABSORB	45s	The "rule" appears — but presented as a <i>recipe</i> or <i>formula card</i> , never a paragraph	Formula card with copy-able pattern	Verbal encoding
5. PRACTICE	90s	4 mini-exercises (mix of types from Section 11)	Interactive widgets	Generation effect
6. MASTERY	30s	Quick self-rated confidence: 😞 / 😊 / 🤖 — feeds SRS algorithm	Three-emoji selector with haptic	Metacognition
7. CELEBRATE	10s	Lottie animation: forge spark, XP counter ticks up, skill node lights up on graph	Particle effect overlay	Dopamine loop

5. VISUAL LANGUAGE SYSTEM — How Grammar is Drawn

This is what separates LinguaForge from every other app. **A consistent visual grammar of grammar.**

5.1 The Visual Vocabulary (Universal Symbols Across All Lessons)

Concept	Visual Symbol	Color	Behavior
Subject (Doer)	☐Purple capsule with person icon	#7C5CFF Royal Iris	Always on left, always elevated
Verb (Action)	☐Cyan diamond with motion lines	#3DDBD9 Mint Cyan	Pulses when active
Object (Receiver)	☐Coral capsule with target icon	#FF7A6B Coral	Receives glowing arrow
Time Marker	☐Gold clock	#D4AF7A Champagne	Floats on timeline
Auxiliary (Helper)	◦ Silver mini-capsule	#A4ADBA Steel	Attaches to main verb
Modifier	☐Translucent glass tile	Glass	Hovers near what it modifies
Negation	☐Red strike-through arc	#FF5577 Danger	Animates as a cross-out swipe
Question Inversion	☐Curved swap arrow	#7C5CFF + animation	Two elements swap positions
Time Flow	➡ Glowing timeline	Aurora gradient	Left-to-right always = past-to-future

5.2 The Five Signature Animations

Animation Name	Purpose	Tech	When Used
The Origami Unfold	New tense reveal	Rive + custom painter	First time a tense is taught
The River of Time	Tense placement on timeline	Flutter Custom Painter + Lottie	Every tense lesson
The Camera Flip	Active → Passive transformation	Rive 3D rotation	Voice lesson
The Branching Universe	Conditionals (real vs hypothetical worlds)	Custom GL shader	Conditionals lessons
The Forge Spark	Mastery achieved	Lottie particle system	Lesson completion

6. THE TENSE VISUALIZATION ENGINE (Signature Feature)

The crown jewel. This is what users will screenshot and share. This is what makes them say "oh my god, I finally GET it."

6.1 The River of Time — Universal Tense Canvas

Every tense lesson opens with the **same canvas** so the brain builds a mental map:

← PAST ——— NOW ——— FUTURE →

▲

(PIN)

[Timeline glows in aurora gradient. 'NOW' has a pulsing dot.]

Onto this canvas, each tense **draws a unique shape**:

Tense	Visual Shape on Timeline	Workplace Example
Simple Past	□ Single dot in the past	"I deployed the build."
Past Continuous	— Glowing line in the past	"I was deploying when Slack went down."
Past Perfect	□ → □ Two dots, arrow between them	"I had deployed before the demo started."
Past Perfect Continuous	— → □ Line ending at past dot	"I had been deploying for 3 hours when it failed."
Simple Present	□ Dot pulsing at NOW (or eternal)	"Our service handles 10K requests/sec."
Present Continuous	— Line covering NOW	"I am deploying the patch."
Present Perfect	□ ← Arrow from past, landing at NOW	"I have deployed three times today."
Present Perfect Continuous	— ← Line from past, landing at NOW	"I have been debugging since morning."
Simple Future	□ Dot in future + arrow from NOW	"I will deploy at 5pm."
Future Continuous	— Line in future	"I will be deploying during lunch."
Future Perfect	□ → □ Future dot, then later dot	"I will have deployed by EOD."
Future Perfect Continuous	— → □ Line ending at future dot	"By Friday I will have been working for 60 hours."

Result: Users mentally see tense as **shape on a timeline**, not as a memorized name.

6.2 The Tense Map — Mastery View

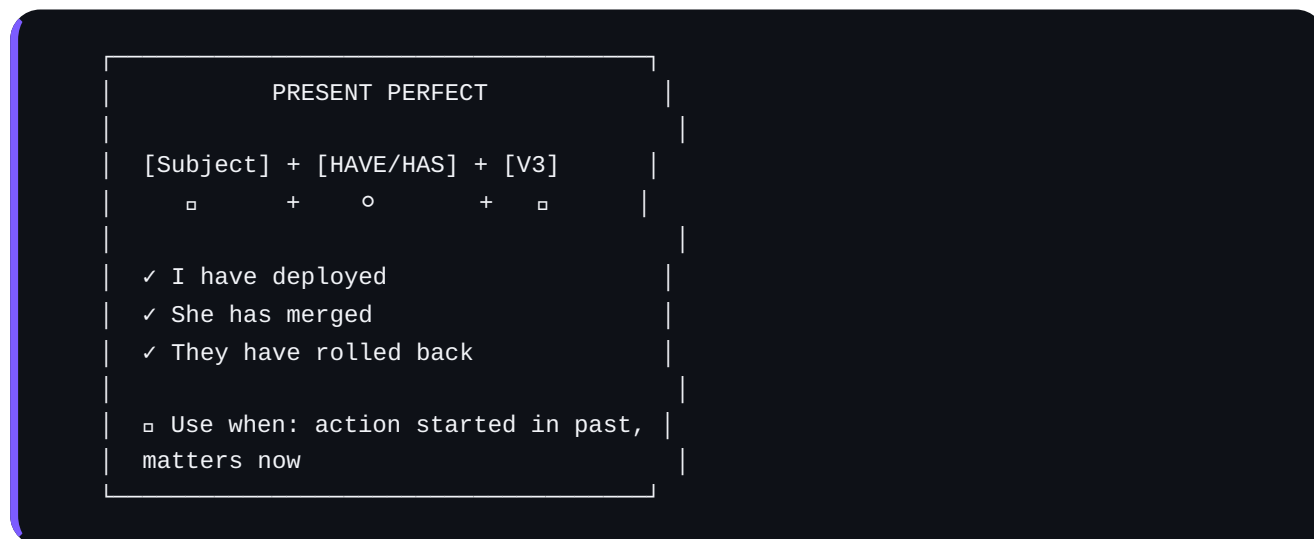
A single 12-cell grid showing all tenses:

	SIMPLE	CONTINUOUS	PERFECT	PERF. CONT.
PAST	[•]	[—]	[• → •]	[— → •]
PRESENT	[•]	[—]	[• → •]	[— → •]
FUTURE	[•]	[—]	[• → •]	[— → •]

Each cell shows the user's **mastery percentage** as a glowing fill. Locked cells are dimmed. Tapping a cell launches that lesson. **This becomes the user's home base for Grammar.**

6.3 The Formula Card (Stage 4 of every lesson)

Instead of paragraph rules, every tense gets a **swipeable formula card**:



User can **swipe up** to see more examples, **swipe down** to see common mistakes.

7. THE SENTENCE SURGERY SYSTEM (Parts of Speech)

For teaching subject/verb/object/modifiers — we use a **dissectable sentence visualization**.

7.1 The Surgery Table

A sentence appears on a virtual operating table. User can: - **Tap any word** → it lifts up with a glow, showing its part of speech - **Drag a word** → see where it can/cannot legally go - **Long-press a word** → swap it with synonyms or alternatives - **Pinch out** → see the syntax tree visualization

7.2 Example Interaction

Original: "The senior engineer quickly merged the pull request."
 ↓ user taps "quickly"

```
ADVERB
Modifies: "merged" (verb)
Can move to: 3 valid positions

Try moving it:
[○] The senior engineer merged
    the PR quickly.
[○] Quickly, the senior engineer
    merged the PR.
```

This is the **tactile dissection** — grammar becomes 3D LEGO, not 2D text.

8. ACTIVE vs PASSIVE VOICE — The Camera Flip Metaphor

8.1 The Concept

A sentence is a **camera shot**. - **Active voice** = camera on the doer - **Passive voice** = camera on the receiver

8.2 The Visualization

Two 3D character avatars on screen: □Subject and □Object.

Active: "**The developer** deployed **the patch**." → camera faces the developer. **Passive:** "**The patch** was deployed **by the developer**." → camera flips 180°, now facing the patch.

User sees an **actual 3D camera rotation animation** between the two voices. Once they see it, they never forget which is which.

8.3 When to Use Which — The Register Compass

A small compass widget shows when each voice fits:

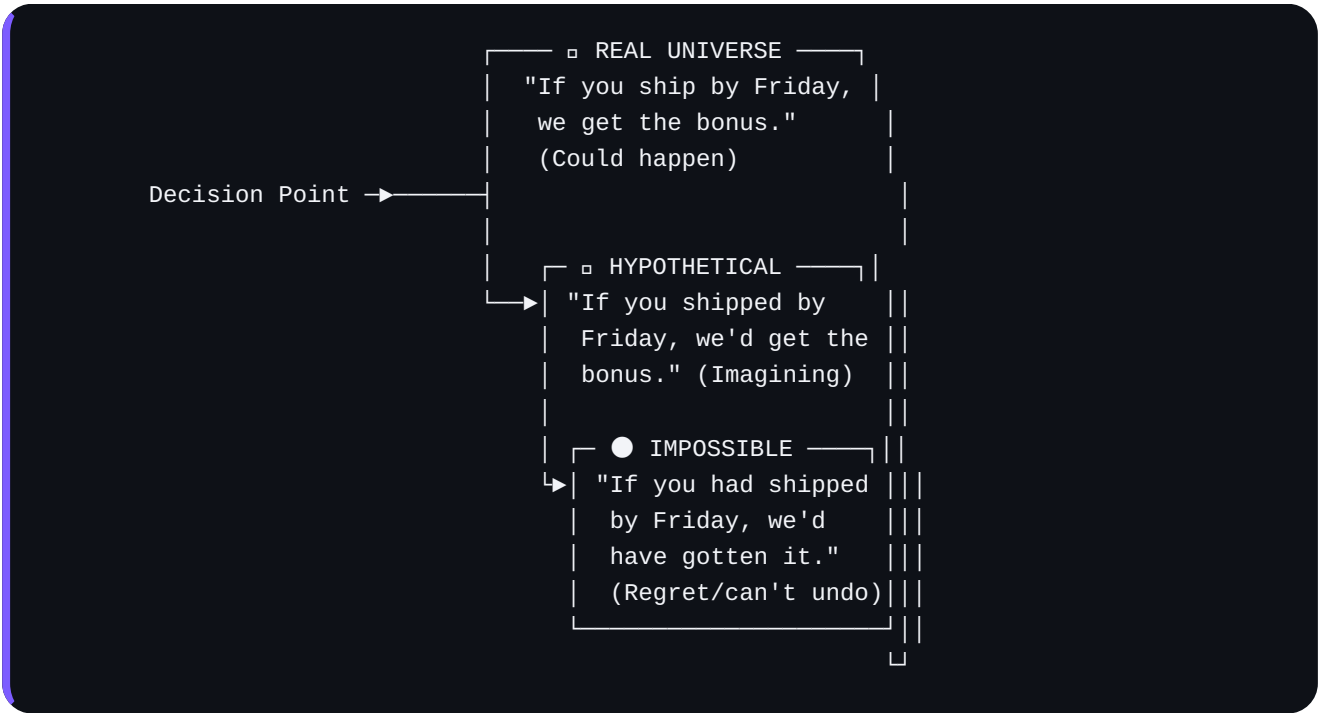
Voice	Best For (Tech Workplace)
Active	Standups, Slack, PR descriptions, demos
Passive	Documentation, post-mortems (depersonalized), academic papers, formal reports

9. CONDITIONALS — The Branching Universe

The hardest grammar topic for Indian/Asian English learners — solved with a **multiverse visualization**.

9.1 The Concept

A conditional is a **fork in reality**. We show it as branching parallel universes.



Each universe gets its own **visual saturation level**: - Real → bright colors, sharp - Hypothetical → soft pastel - Impossible (past) → desaturated, ghost-like

Users **swipe between universes** to feel the shift in probability.

9.2 The 5 Conditional Types Mapped

Type	Universe	Pattern	Tech Example
Zero	Universal truth	If + V1, V1	"If the build fails, CI alerts us."
First	Real-possible	If + V1, will + V1	"If the test passes, we'll merge."
Second	Imaginary-now	If + V2, would + V1	"If I were CTO, I'd refactor this."
Third	Impossible-past	If + had + V3, would have + V3	"If we had load-tested, it wouldn't have crashed."
Mixed	Past affecting now	If + had + V3, would + V1	"If we had hired earlier, we'd be shipping now."

10. ADAPTIVE DIFFICULTY & AI TUTOR

10.1 The Adaptive Engine

A **Bayesian Knowledge Tracing (BKT)** model running on-device tracks: - **P(knows)** — probability user knows the concept - **P(guess)** — probability of correct answer despite not knowing - **P(slip)** — probability of mistake despite knowing - **P(transit)** — probability of learning from this exercise

After each answer, all 4 update. When **P(knows) > 0.95** for 3 consecutive sessions → concept **mastered** → graduate to next.

10.2 The AI Tutor — "Why am I wrong?"

When user gets something wrong, a small floating button appears: **"Ask Maya"** (the AI tutor persona).

Maya uses **Gemini 2.0 Flash** with this system prompt:

You are Maya, a senior English tutor for tech professionals.
 The user wrote: "{user_answer}"
 The correct answer was: "{correct_answer}"
 The grammar concept: "{concept}"

Respond in EXACTLY this format:

1. One-line empathetic acknowledgment (no "great question" filler)
2. Show the difference in a code-block format:
 - {user_answer}
 - {correct_answer}
3. Explain WHY in 2 sentences, using a workplace analogy
4. Give 1 mini-exercise to lock it in

Tone: peer-level, warm, never patronizing. Max 80 words total.

10.3 The Skill Graph (User's Personal Map)

A force-directed graph visualization shows the user's grammar knowledge as **constellations**: - Each concept = a glowing node - Mastered concepts = bright, large, gold - Wobbly concepts = pulsing, amber - Locked concepts = dim, gray - Lines connect prerequisite relationships

Users see their **knowledge growing in real time**.

11. INTERACTIVE EXERCISE LIBRARY (14 Types)

#	Exercise	Mechanic	Cognitive Demand	Used In
1	Tap the Difference	Two sentences, tap what changed	Recognition	Hook (Stage 2)
2	Drag-to-Build	Drag word tiles into correct order	Reconstruction	All tenses
3	Fill the Gap	Tap or type missing word	Recall	All modules
4	Conjugate Race	Verb conjugation against timer	Speed automaticity	Tenses
5	Sentence Surgery	Tap a word, change its role	Syntactic awareness	Parts of speech
6	Voice Flip	Convert active ↔ passive	Transformation	Voice module
7	Universe Hopper	Switch conditional types	Modality	Conditionals
8	Speak the Sentence	Voice recording + AI scoring	Production	All (optional)
9	Spot the Bug	Find grammatical error (like code review!)	Error detection	Mixed review
10	Translate from Hinglish	Convert "I am having lunch" → standard	Code-switching	Articles, tenses
11	Email Builder	Assemble a professional email from chunks	Register	Connectors
12	Standup Generator	Generate yesterday/today/ blocker statements	Production	Tenses (Past, Present, Future)
13	Code Review Phrasing	Pick the politest valid sentence	Pragmatics	Modals
14	PR Description Polish	Rewrite a sloppy PR description	Holistic	End of module

Each exercise has: - ☐ Haptic feedback (light tap on correct, double tap on wrong) - ☐ Sound effect (subtle — forge spark on right, soft chime on wrong) - ☐ Visual: green glow on right, coral pulse on wrong - ☐ Instant explanation tap (never wait)

12. MISTAKE FORENSICS & PERSONAL WEAKNESS MAP

Every wrong answer is **logged with context**:

```
{
  "userId": "u_47291",
  "lessonId": "tenses_present_perfect",
  "exerciseId": "ex_073",
  "userAnswer": "I have deploy the build",
  "correctAnswer": "I have deployed the build",
  "errorType": "MISSING_PARTICIPLE",
  "concept": "present_perfect_form",
  "timestamp": "2026-06-03T14:22:00Z",
  "timeToAnswer": 4.2,
  "confidence": "medium"
}
```

12.1 Pattern Detection (Weekly Job)

A Gemini-powered pipeline analyzes the mistake log and surfaces patterns:

"Arjun — you've made 7 similar mistakes this week. You consistently drop -ed endings in Present Perfect. Let's spend 2 minutes on this."

A **Weakness Card** appears on the home screen. Tapping it launches a 2-minute targeted micro-lesson.

12.2 The Heat Map View

User can view their mistakes as a **heat map across the 12 modules** — red zones = focus areas, green = strong. Pure clarity. No shame, no judgment.

13. GAMIFICATION & MASTERY SYSTEM

13.1 The Forge Metaphor (Brand-Native)

Every concept mastered = one **strike on the anvil**. Users are *forging* their English.

Element	Mechanic
XP	Earned per lesson, per correct streak, per mastery
Forge Sparks	Currency — earned, spent on hint/skip
Anvil Levels	1–10 mastery levels per module, with visual upgrade of an anvil 3D icon
Badges	"Tense Master", "Article Slayer", "Conditional Conqueror"
Streaks	Daily grammar practice streak — separate from app streak
Leaderboard	Optional, only among consented peers — never global

13.2 The Mastery Ladder

Each grammar module has 5 ranks: **Apprentice** → **Smith** → **Forger** → **Master** → **Grandmaster** Each rank unlocks new exercise difficulty + bonus content (real PR examples, interview clips).

14. TECHNICAL ARCHITECTURE — Flutter Implementation

14.1 Folder Structure

```

lib/features/grammar/
├── data/
│   ├── models/
│   │   ├── lesson.dart
│   │   ├── tense_definition.dart
│   │   ├── exercise.dart
│   │   ├── mistake.dart
│   │   └── mastery_state.dart
│   ├── repositories/
│   │   ├── grammar_repository.dart
│   │   └── mistake_repository.dart
│   └── sources/
│       ├── grammar_content_local.dart    # Bundled JSON
│       ├── grammar_content_remote.dart   # Firestore
│       └── srs_engine.dart                # SuperMemo SM-2
├── domain/
│   ├── entities/
│   ├── usecases/
│   │   ├── start_lesson.dart
│   │   ├── submit_answer.dart
│   │   ├── compute_next_review.dart
│   │   └── analyze_mistakes.dart
│   └── adaptive/
│       └── bkt_engine.dart                # Bayesian Knowledge Tracing
├── presentation/
│   ├── screens/
│   │   ├── grammar_hub_screen.dart       # The 12-cell tense map
│   │   ├── lesson_player_screen.dart     # The 7-stage flow
│   │   ├── exercise_screens/             # 14 exercise widgets
│   │   ├── mistake_analyzer_screen.dart
│   │   └── skill_graph_screen.dart
│   ├── widgets/
│   │   ├── river_of_time.dart            # Custom painter timeline
│   │   ├── formula_card.dart
│   │   ├── sentence_surgery_table.dart
│   │   ├── camera_flip_widget.dart       # Rive integration
│   │   ├── universe_hopper.dart          # Conditionals
│   │   └── forge_anvil.dart              # Mastery indicator
│   └── animations/
│       ├── origami_unfold.dart
│       └── forge_spark.dart
└── providers/                            # Riverpod
    ├── lesson_provider.dart
    ├── mastery_provider.dart
    └── ai_tutor_provider.dart

```

14.2 Core Technologies

Layer	Tech	Why
State	Riverpod 2.x	Compile-safe, async-first
Animation	Rive + Flutter Animate + Custom Painter	Rive for 3D, Painter for timelines
Local DB	Hive (encrypted)	Offline lessons + mistake log
Sync	Firestore	Cross-device progress
Voice	Groq Whisper v3	Speech exercise scoring
AI	Gemini 2.0 Flash	Tutor, mistake analysis
Charts	fl_chart + custom painters	Skill graph, heat map

14.3 Performance Budget (Non-Negotiable)

- **Cold start to lesson:** $\leq 1.2s$
- **Animation frame rate:** 60 fps minimum (120 on ProMotion)
- **AI tutor response:** $\leq 1.8s$ p95
- **Offline-first:** all 88 lessons usable without network

15. AI PROVIDER ROUTING

Smart routing based on task — keeping within free tiers:

Task	Provider	Model	Why
Lesson explanation generation	Gemini	2.0 Flash	Free, fast, accurate
"Why am I wrong?" tutor	Gemini	2.0 Flash	Conversational, concise
Mistake pattern analysis (weekly)	Gemini	2.0 Pro	Long context, batch job
Sentence generation (examples)	Groq	Llama 3.3 70B	Sub-second
Speech-to-text (speak exercises)	Groq	Whisper Large v3	Free, fast
Pronunciation scoring	AWS Bedrock	Claude Haiku	Reliability
Fallback / overflow	OpenRouter	Mistral 7B	Cheap backup
Content moderation	Gemini	Safety filter	Built-in

A **GrammarAIRouter** class abstracts this — code never calls providers directly.

16. ACCESSIBILITY & INCLUSION

Feature	Implementation
Dyslexia mode	OpenDyslexic font option, increased letter spacing
Reduced motion	Animations replaced with cross-fades when <code>MediaQuery.disableAnimations</code>
Screen reader	Every grammar visual has semantic label (e.g., "Timeline showing past perfect tense with arrow from past to past")
Large text	All UI scales to 1.5× without breaking
Color-blind safe	No information conveyed by color alone — always paired with shape/icon
Hinglish bridge	Optional toggle: "Show in Hindi" appears for hard concepts
One-handed mode	All controls reachable from bottom 60% of screen
Offline mode	Full functionality without network (except AI tutor)

17. CONTENT PRODUCTION PIPELINE

17.1 Roles

- **Lead Linguist** writes lesson scripts
- **Tech-Domain SME** validates workplace examples
- **Motion Designer** creates Rive files
- **Content Engineer** converts to structured JSON
- **QA Linguist** reviews + signs off

17.2 Lesson JSON Schema

```
{
  "id": "present_perfect_v1",
  "module": "tenses",
  "title": "Present Perfect – Bridging Past & Now",
  "difficulty": 3,
  "estimatedMinutes": 5,
  "prerequisites": ["simple_past_v1", "past_participles_v1"],
  "stages": {
    "hook": { "scenario": "...", "asset": "rive/hook_pp.riv" },
    "discover": { "sentences": [...], "diff": [...] },
    "reveal": { "animation": "rive/timeline_pp.riv", "narration": "..." },
    "absorb": { "formula": "Subject + HAVE/HAS + V3", "examples": [...] },
    "practice": { "exercises": ["ex_001", "ex_002", "ex_003", "ex_004"] },
    "mastery": { "selfRate": true }
  },
  "tags": ["tech", "high-frequency", "interview-critical"]
}
```

17.3 Content Volume — Launch Bar

- 88 lessons fully written + visually animated
- 1,200+ exercises across 14 types
- 500+ tech-workplace example sentences
- 12 Rive animations (one per module)
- 88 Lottie celebration variants

18. SUCCESS METRICS & KPIs

Metric	Target	Measured Via
D1 retention (grammar users)	≥ 55%	Firebase Analytics
D7 retention	≥ 32%	Firebase Analytics
Lessons completed per user / week	≥ 3.5	Firestore aggregation
Mastery achieved (any module)	≥ 40% of users in 30 days	BKT events
Mistake reduction over 4 weeks	≥ 35% per concept	Mistake log delta
NPS — "Grammar Engine"	≥ 60	In-app survey
Avg session length	8–12 min (sweet spot)	Session events
Animation FPS p95	≥ 58 fps	Firebase Perf Monitoring

19. DEVELOPMENT ROADMAP

Sprint Plan (2-week sprints)

Sprint	Focus	Deliverable
Sprint 1	Foundation	Grammar Hub screen + 12-cell tense map UI
Sprint 2	Lesson Player	7-stage shell + Stage 1–3 animations
Sprint 3	Visual Theatre	River of Time + Origami Unfold (3 tenses pilot)
Sprint 4	Practice Lab v1	6 of 14 exercise types
Sprint 5	Adaptive Engine	BKT + SRS implementation
Sprint 6	AI Tutor	Maya integration + mistake logging
Sprint 7	Remaining Tenses	9 more tenses + Camera Flip + Branching Universe
Sprint 8	Practice Lab v2	8 remaining exercise types
Sprint 9	Other Modules	Articles, Prepositions, Modals (45 lessons)
Sprint 10	Other Modules	Voice, Conditionals, Reported Speech (19 lessons)
Sprint 11	Other Modules	Phrasal Verbs, Connectors, Questions, Punctuation (23 lessons)
Sprint 12	Polish & Beta	Performance, a11y, beta launch

Total: 24 weeks (~6 months) to v1.0 GA. Beta after Sprint 8 (~4 months).

20. RISK REGISTER

Risk	Likelihood	Impact	Mitigation
Rive animations slow on low-end Android	Medium	High	Provide Lottie fallback, detect device tier
AI free-tier rate limits hit	Medium	Medium	Multi-provider routing, queue + cache
Content production becomes bottleneck	High	High	Hire 2 content engineers, build authoring tool
Users skip "boring" Discover stage	Medium	Medium	A/B test, make Discover more game-like
Indian English vs Standard English friction	Medium	High	Add "Indian English Mode" toggle (no judgment)
3D performance kills battery	Low	High	Render budget cap; auto-degrade after 30 min
Gemini API changes break tutor	Medium	Medium	OpenRouter as live fallback
Mistake log creates anxiety	Medium	Medium	Frame as "growth zone", never "weakness"

□ ACCEPTANCE CRITERIA — When Grammar Engine is "Done"

- [] All 12 modules content-complete with 88 lessons
- [] All 14 exercise types implemented and play-tested
- [] All 5 signature animations (Origami, River, Camera Flip, Universe, Forge) production-grade
- [] BKT + SRS engine running, validated against 500 user sessions
- [] AI Tutor (Maya) responds in <1.8s p95 with quality bar
- [] Mistake forensics surface accurate weakness cards weekly
- [] Accessibility: passes WCAG AA, dyslexia mode tested with users
- [] Performance: 58+ fps p95 on mid-tier Android (Pixel 6a baseline)
- [] Beta NPS ≥ 50 from 500+ users
- [] Linguistic accuracy: 0 errors flagged by external CELTA review

□ THE NORTH STAR

If after using LinguaForge Grammar Engine for 30 days, a user can:

1. **Confidently speak about deploys, sprints, and incidents** in 3 tenses without thinking
2. **Catch their own grammar mistakes** in Slack/email before sending
3. **Pass a tech interview's grammar bar** without preparation
4. **Feel pride, not anxiety, about their English**

— then we have built the right thing.

"We're not teaching grammar. We're forging confident communicators." — Project Origami, LinguaForge

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